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Project 2

Automation of Concrete Mix Design

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Introduction

Project Background

The Nebraska Department of Transportation (NDOT) currently relies on an Excel-based workflow to perform concrete mix design calculations. While the existing spreadsheet accurately implements the required engineering logic, NDOT identified several limitations— including difficulty maintaining formulas, limited transparency, and challenges integrating the process into modern digital tools. To improve reliability and streamline future updates, NDOT contracted our team to convert the Excel mix design procedure into an automated Python-based system.

This project focuses on translating the full mix-design workflow from Excel into clearly defined Python functions that accept user inputs such as cement content, water–cement ratio, air content, and aggregate proportions. The automated program organizes the calculations, generates a professional weight chart for one cubic yard of concrete, and maintains consistency with the original NDOT design methodology.

Beyond simple conversion, the project improves usability, creates a more robust structure for future expansion, and enables scenario testing through four realistic concrete mix cases. These enhancements provide NDOT with a reliable, transparent, and repeatable tool suitable for future web-based deployment

Scope of Work

This project concentrates on converting the Nebraska Department of Transportation (NDOT) Concrete Mix Design Excel worksheet into a fully automated Python-based system. The goal is to translate all engineering calculations from the Excel Mix Design tab while improving transparency, consistency, and ease of use. The automated workflow will guide users through entering material properties and mix parameters, then generate a professional weight summary for one cubic yard of concrete.

To accomplish this, the team first analyzed the Excel workbook to understand the underlying engineering logic, including absolute volume calculations, moisture corrections, and yield checks. All relevant formulas were documented and translated into modular Python functions. A structured input workflow was created to ensure users provide values in a logical engineering sequence, with clear units and validation checks.

The program's output will present organized material quantities—cementitious components, water, air, fine aggregate, and coarse aggregate—along with a yield verification to ensure mix accuracy. To validate the model, four realistic concrete mix scenarios (e.g., pavement, bridge deck, structural, high-strength) will be researched, entered into the automated system, and evaluated for consistency.

Final deliverables include: a functioning Jupyter Notebook, annotated code documentation, a GitHub repository with a user guide, a Gantt chart, the Scope of Work, and a comprehensive technical report.

This automated system is expected to reduce user error, improve calculation transparency, and serve as a foundation for future web-based engineering tools.

Method

Coding

To begin, assign the fundamental constant values used for calculations throughout the code to variables. These constants are the unit weight of water, which is 62.4 lb/ft^3, and the cubic feet in one cubic yard, which is 27 ft^3. Then begin to replicate the calculations from the Mix Design tab of the excel file in code. Each calculation will be defined as a new function. To calculate water weight, define a function with a formula that adds the weights of cement, fly ash, silica fume, and other SCM and then multiply that sum by the target water/cement ratio. To calculate the volume of cement, define a function with a formula that will take the unit weight of water multiplied by the specific gravity of cement and divide this product from the weight of cement. To calculate the volume of fly ash, define a function with a formula that will take the unit weight of water multiplied by the specific gravity of fly ash and divide this product from the weight of fly ash. To calculate the volume of silica fume, define a function with a formula that will take the unit weight of water multiplied by the specific gravity of silica fume and divide this product from the weight of silica fume. To calculate the volume of other SCM, define a function with a formula that will take the unit weight of water multiplied by the specific gravity of other SCM and divide this product from the weight of other SCM. To calculate the volume of air, define a function with a formula that will divide the target percentage of air content by 100, then multiplying this quotient by the constant of cubic feet in one yard. To calculate the volume of water, define a function with a formula to divide the water weight by the weight of water constant. To calculate the volume of the total aggregate, define a function to subtract the volumes of

cement, fly ash, silica fume, other SCM, air, and water from the constant of cubic feet in one yard. To calculate the weight of fine aggregate, define a function with a formula that divides the percentage of fine aggregate by 100, them multiplies that quotient by the weight of water constant, the specific gravity of fine aggregate, and the volume of total aggregate. To calculate the weight of coarse aggregate, define a function with a formula that divides the percentage of coarse aggregate by 100, them multiplies that quotient by the weight of water constant, the specific gravity of coarse aggregate, and the volume of total aggregate. To calculate the weight of other aggregates, define a function with a formula that divides the percentage of other aggregates by 100, then multiplies that quotient by the weight of water constant, the specific gravity of other aggregate, and the volume of total aggregate.

Next input functions need to be coded and assigned to variables so that the values input by the user can be put into the formulas previously defined. User inputs needed include: project number, class of concrete, weights of cement, fly ash, silica fume, and other SCM, the target water to cement ratio, percentages of air content, fine aggregate, coarse aggregate, and other aggregate, and specific gravities of cement, fly ash, silica fume, other SCM, fine aggregate, coarse aggregate, and other aggregates.

After that, the final variables will need to be assigned to the results from plugging the user input values into the previously defined equation functions. These variables include the following: water weight, volumes of cement, fly ash, silica fume, other SCM, air, water, total aggregate, fine aggregate, coarse aggregate, and other aggregates.

Finally, use print statements to generate a summary of the findings and results of the user input values. The summary should include a title like “Weight Summary Per Cubic Yard of Concrete”, followed by project number and class of concrete. The summary should then list the following: weights of cement, fly ash, silica fume, other SCM, fine aggregate, coarse aggregate, other aggregate, and water. Conclude the summary with a statement such as: “End of Mix Design Summary”.

Research

The research consisted of finding realistic and practical concrete mix design scenarios to input when running the code. For these scenarios, the values used were taken from Table 1a. Concrete Mix Designs for Bridge Decks Used by State DOTs (Xi, Shing, and Xie 2001). Scenario 1 utilizes data from the New York row of the table, with assumptions for the other SCM weight, percentage of aggregates, and specific gravities inputs, as these values were not included in the data table. Scenario 2 utilizes data from the Nebraska row of the table, with assumptions for the other SCM weight, percentage of aggregates, and specific gravities inputs, as these values were not included in the data table. Scenario 3 utilizes data from the Colorado row of the table, with assumptions for the other SCM weight, percentage of aggregates, and specific gravities inputs, as

these values were not included in the table. Scenario 4 utilizes data from the Texas row of the table, with assumptions for the other SCM weight, percentage of aggregates, and specific gravities inputs, as these values were not included in the table. With each of these input values researched and ready, the code can be run to evaluate each different concrete mix design weight summary.

Table 1a. Concrete Mix Designs for Bridge Decks Used by State DOTs

States	Cement (lb/yd^3)	Fly ash (lb/yd^3)	Silica fume (lb/yd^3)	w/(c+m)	28-d Strength (psi)	Permeability 28d (Coul.)	Air content (%)	Slump (inch)
Colorado Shing, P.B.et al, 1999)	660	-	50	0.35	5800	-	4-8	
Colorado	615-660	<61-66	-	<0.44	4500	-	5-8	
Illinois (Detwiler,1997)	630	-	70	0.31	6950 at 14d	540	6-8	-
New York (Alampalli,2000)	505	149	42	0.4			6.5	3-4
Washington (FHWA-RD-00-124)	660	75	-	0.39	4000 5300 at56d	2800	6.0	-
Nebraska Beacham, M. W. (1999)	750	75	-	0.31	8000 at 56d	589 at 56d	6.0	-
Texas (Ralls, M. L., 1999)	382-610	88-131	-	0.31-0.43	4000	<2000	5-8	3-9
New Hampshire (Waszczuk, C. M. et al, 1999)	607	-	45	0.383	6000 7200 at 56d	<1000 at 56d	6-9	3-5
Virginia (FHWA-RD-00-123)	560	140	-	0.45	5000	2500	-	-

Results

Concrete Mix Design Scenario 1: New York

Using the values from the New York row in Table 1a, the first concrete mix design scenario was calculated. All inputs and the weight summary per cubic foot response are demonstrated in Figure 1, which was generated using the automated mix design code. Out of the four scenarios, this one had the lowest weight of cement and other SCM per cubic foot. It is also the only scenario out of the four that contains silica fume.

```
Enter project number:  1
Enter class of concrete (e.g., 47B, 47BR):  SF
Enter cement weight A (lb per cubic yard):  505
Enter fly ash weight B (lb per cubic yard):  149
Enter silica fume weight C (lb per cubic yard):  42
Enter other SCM weight D (lb per cubic yard):  0
Enter target water-cement ratio E:  0.4
Enter target air content F (%):  6.5
Enter percent fine aggregate G (%):  33
Enter percent coarse aggregate H (%):  33
Enter percent other aggregate I (%):  34
Enter specific gravity of cement J:  3.1
Enter specific gravity of fly ash K:  2.8
Enter specific gravity of silica fume L:  2.2
Enter specific gravity of other SCM M:  3.1
Enter specific gravity of fine aggregate N:  2.62
Enter specific gravity of coarse aggregate O:  2.62
Enter specific gravity of other aggregate P:  2.62

--------------------------------------------
 NDOT Concrete Mix Design - Weight Summary
         (1 Cubic Yard of Concrete)
--------------------------------------------
Project Number:       1
Class of Concrete:    SF
--------------------------------------------
Cement (A):               505.0 lb
Fly Ash (B):              149.0 lb
Silica Fume (C):           42.0 lb
Other SCM (D):              0.0 lb
--------------------------------------------
Fine Aggregate (Y):         918 lb
Coarse Aggregate (Z):       918 lb
Other Aggregate (AA):       946 lb
--------------------------------------------
Water (Q):                  278 lb
--------------------------------------------
End of Mix Design Summary
```

Figure 1 Concrete Mix Design Scenario 1

Concrete Mix Design Scenario 2: Nebraska

Using the values from the Nebraska row in Table 1a, the second concrete mix design scenario was calculated. All inputs and the weight summary per cubic foot response are demonstrated in Figure 2, which was generated using the automated mix design code. Out of the four scenarios, this one had the lowest weight of water per cubic foot. This is reasonable as this scenario also has the lowest target water-cement ratio out of the four. This mix also has the largest total weight out of the four.

```
Enter project number:  2
Enter class of concrete (e.g., 47B, 47BR):  SF
Enter cement weight A (lb per cubic yard):  750
Enter fly ash weight B (lb per cubic yard):  75
Enter silica fume weight C (lb per cubic yard):  0
Enter other SCM weight D (lb per cubic yard):  42
Enter target water-cement ratio E:  0.31
Enter target air content F (%):  6
Enter percent fine aggregate G (%):  20
Enter percent coarse aggregate H (%):  40
Enter percent other aggregate I (%):  40
Enter specific gravity of cement J:  3.1
Enter specific gravity of fly ash K:  2.8
Enter specific gravity of silica fume L:  2.2
Enter specific gravity of other SCM M:  3.1
Enter specific gravity of fine aggregate N:  2.62
Enter specific gravity of coarse aggregate O:  2.62
Enter specific gravity of other aggregate P:  2.62

---------------------------------------------
 NDOT Concrete Mix Design - Weight Summary
         (1 Cubic Yard of Concrete)
---------------------------------------------
Project Number:       2
Class of Concrete:    SF
---------------------------------------------
Cement (A):               750.0 lb
Fly Ash (B):               75.0 lb
Silica Fume (C):            0.0 lb
Other SCM (D):             42.0 lb
---------------------------------------------
Fine Aggregate (Y):         541 lb
Coarse Aggregate (Z):      1082 lb
Other Aggregate (AA):      1082 lb
---------------------------------------------
Water (Q):                  269 lb
---------------------------------------------
End of Mix Design Summary
```

Figure 2 Concrete Mix Design Scenario 2

Concrete Mix Design Scenario 3: Colorado

Using the values from the Colorado row in Table 1a, the third concrete mix design scenario was calculated. All inputs and the weight summary per cubic foot response are demonstrated in Figure 3, which was generated using the automated mix design code. Out of the four scenarios, this one had the lowest weight of fly ash per cubic foot.

```
Enter project number:  3
Enter class of concrete (e.g., 47B, 47BR):  SF
Enter cement weight A (lb per cubic yard):  630
Enter fly ash weight B (lb per cubic yard):  63
Enter silica fume weight C (lb per cubic yard):  0
Enter other SCM weight D (lb per cubic yard):  42
Enter target water-cement ratio E:  0.43
Enter target air content F (%):  7
Enter percent fine aggregate G (%):  33
Enter percent coarse aggregate H (%):  33
Enter percent other aggregate I (%):  34
Enter specific gravity of cement J:  3.1
Enter specific gravity of fly ash K:  2.8
Enter specific gravity of silica fume L:  2.2
Enter specific gravity of other SCM M:  3.1
Enter specific gravity of fine aggregate N:  2.62
Enter specific gravity of coarse aggregate O:  2.62
Enter specific gravity of other aggregate P:  2.62

--------------------------------------------
 NDOT Concrete Mix Design - Weight Summary
         (1 Cubic Yard of Concrete)
--------------------------------------------
Project Number:       3
Class of Concrete:    SF
--------------------------------------------
Cement (A):               630.0 lb
Fly Ash (B):               63.0 lb
Silica Fume (C):            0.0 lb
Other SCM (D):             42.0 lb
--------------------------------------------
Fine Aggregate (Y):         875 lb
Coarse Aggregate (Z):       875 lb
Other Aggregate (AA):       901 lb
--------------------------------------------
Water (Q):                  316 lb
--------------------------------------------
End of Mix Design Summary
```

Figure 3 Concrete Mix Design Scenario 3

Concrete Mix Design Scenario 4: Texas

Using the values from the Texas row in Table 1a, the fourth concrete mix design scenario was calculated. All inputs and the weight summary per cubic foot response are demonstrated in Figure 4, which was generated using the automated mix design code. Out of the four scenarios, this one had the highest weight of water per cubic foot and the lowest total weight.

```
Enter project number:  4
Enter class of concrete (e.g., 47B, 47BR):  SF
Enter cement weight A (lb per cubic yard):  608
Enter fly ash weight B (lb per cubic yard):  120
Enter silica fume weight C (lb per cubic yard):  0
Enter other SCM weight D (lb per cubic yard):  42
Enter target water-cement ratio E:  0.42
Enter target air content F (%):  6
Enter percent fine aggregate G (%):  20
Enter percent coarse aggregate H (%):  40
Enter percent other aggregate I (%):  40
Enter specific gravity of cement J:  3.1
Enter specific gravity of fly ash K:  2.8
Enter specific gravity of silica fume L:  2.2
Enter specific gravity of other SCM M:  3.1
Enter specific gravity of fine aggregate N:  2.62
Enter specific gravity of coarse aggregate O:  2.62
Enter specific gravity of other aggregate P:  2.62

--------------------------------------------
 NDOT Concrete Mix Design - Weight Summary
         (1 Cubic Yard of Concrete)
--------------------------------------------
Project Number:       4
Class of Concrete:    SF
--------------------------------------------
Cement (A):               608.0 lb
Fly Ash (B):              120.0 lb
Silica Fume (C):            0.0 lb
Other SCM (D):             42.0 lb
--------------------------------------------
Fine Aggregate (Y):         528 lb
Coarse Aggregate (Z):      1056 lb
Other Aggregate (AA):      1056 lb
--------------------------------------------
Water (Q):                  323 lb
--------------------------------------------
End of Mix Design Summary
```

Figure 4 Concrete Mix Design Scenario 4

References

1.Xi, Y., Shing, B. and Xie, Z. https://hermes.cde.state.co.us/islandora/object/co%3A5071/datastream/OBJ/download/Development_of_optimal_concrete_mix_designs_for_bridge_decks.pdf (accessed Feb 26, 2026).

2.Mix Design Submittal https://dot.nebraska.gov/media/jp3paote/mix-design-submittal.xlsx (accessed Feb 26, 2026).